Task: 1. Conceptual Model (Entity-Relationship)

The conceptual model identifies the core entities, their attributes, and the relationships between them. The diagram below represents the full ERD for the system.

1.1 Entities

PROVIDER

Represents a registered business or individual that offers products or services on the platform.

Attribute	Data Type	Constraints / Notes
provider_id (PK)	Integer	Primary key, auto-increment, NOT NULL
business_name	VARCHAR(200)	NOT NULL
email	VARCHAR(200)	UNIQUE, NOT NULL
phone	VARCHAR(30)	Optional
address	TEXT	Optional
status	ENUM	Values: active, suspended, inactive
registered_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

PRODUCT_SERVICE

Represents any product or service offered by a provider. Includes full catalog details such as pricing, availability, and specifications.

Attribute	Data Type	Constraints / Notes
product_id (PK)	Integer	Primary key, auto-increment, NOT NULL
provider_id (FK)	Integer	FK references PROVIDER(provider_id), NOT NULL
name	VARCHAR(200)	NOT NULL
category	VARCHAR(100)	NOT NULL
price	DECIMAL(12,2)	NOT NULL, >= 0
availability	ENUM	Values: in_stock, out_of_stock, service_available
stock_quantity	INTEGER	NULL allowed for services
specifications	TEXT	Optional descriptive details
updated_at	TIMESTAMP	Auto-updated on change

CONSUMER

Represents a registered end user who browses the platform and places reservations.

Attribute	Data Type	Constraints / Notes
consumer_id (PK)	Integer	Primary key, auto-increment, NOT NULL
full_name	VARCHAR(200)	NOT NULL
email	VARCHAR(200)	UNIQUE, NOT NULL
phone	VARCHAR(30)	Optional
address	TEXT	Optional
registered_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

PLATFORM_CHANNEL

Represents the digital channel through which a reservation was made (e.g., website, social media platform, mobile app).

Attribute	Data Type	Constraints / Notes
channel_id (PK)	Integer	Primary key, auto-increment, NOT NULL
channel_name	VARCHAR(100)	NOT NULL
channel_type	ENUM	Values: website, social_media, mobile_app
url	VARCHAR(300)	Optional

RESERVATION

Captures the consumer's order request. A reservation must have a consumer and at least one product item. It cannot exist independently of both.

Attribute	Data Type	Constraints / Notes
reservation_id (PK)	Integer	Primary key, auto-increment, NOT NULL
consumer_id (FK)	Integer	FK references CONSUMER(consumer_id), NOT NULL
channel_id (FK)	Integer	FK references PLATFORM_CHANNEL(channel_id), nullable
reservation_date	DATE	NOT NULL
status	ENUM	Values: pending, confirmed, cancelled, fulfilled

Attribute	Data Type	Constraints / Notes
notes	TEXT	Optional
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

RESERVATION_ITEM

A bridge/associative entity that links reservations to specific products. Each row is one line item. Unit price is stored as a snapshot at time of reservation to preserve historical accuracy.

Attribute	Data Type	Constraints / Notes
item_id (PK)	Integer	Primary key, auto-increment, NOT NULL
reservation_id (FK)	Integer	FK references RESERVATION(reservation_id), NOT NULL
product_id (FK)	Integer	FK references PRODUCT_SERVICE(product_id), NOT NULL
quantity	INTEGER	NOT NULL, >= 1
unit_price	DECIMAL(12,2)	Snapshot of price at reservation time, NOT NULL

PAYMENT

Records the payment associated with a reservation. The 1:1 relationship with RESERVATION is enforced via a unique constraint on reservation_id.

Attribute	Data Type	Constraints / Notes
payment_id (PK)	Integer	Primary key, auto-increment, NOT NULL
reservation_id (FK)	Integer	FK + UNIQUE references RESERVATION(reservation_id), NOT NULL
payment_method	ENUM	Values: cheque, electronic_card
amount	DECIMAL(12,2)	NOT NULL, >= 0
status	ENUM	Values: pending, completed, failed, refunded
payment_timestamp	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

TRANSACTION

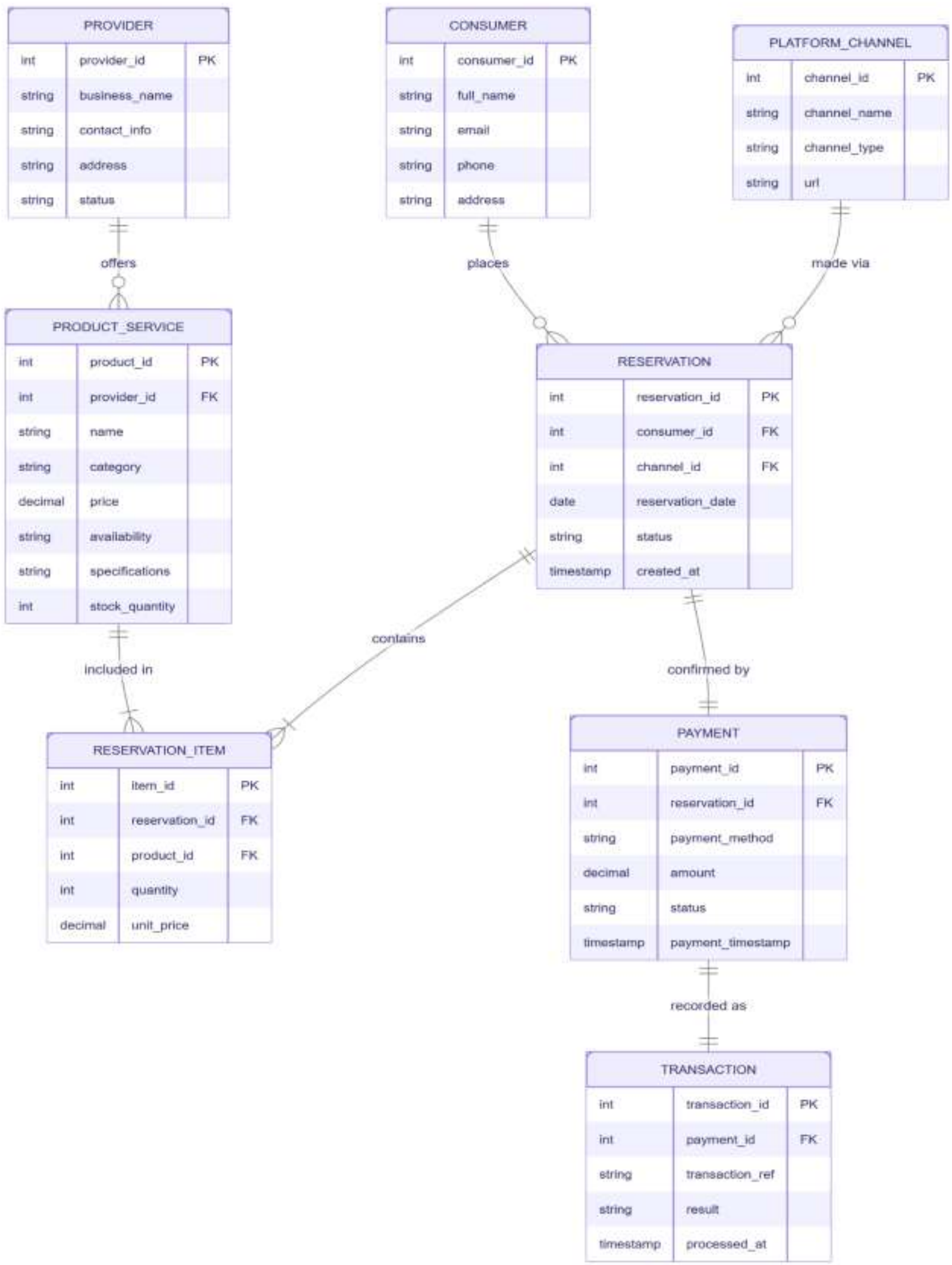
Records the technical result of the payment processing attempt. Separate from PAYMENT to support retries and failure tracking without overwriting payment intent data.

Attribute	Data Type	Constraints / Notes
transaction_id (PK)	Integer	Primary key, auto-increment, NOT NULL
payment_id (FK)	Integer	FK + UNIQUE references PAYMENT(payment_id), NOT NULL
transaction_ref	VARCHAR(100)	UNIQUE, external gateway reference
result	ENUM	Values: success, failed, timeout
failure_reason	TEXT	Populated on failure, optional
processed_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

1.2 Relationship Summary

The following table summarizes all identified relationships between entities in the conceptual model.

Entity A	Entity B	Cardinality	Description
PROVIDER	PRODUCT_SERVICE	1 : many	A provider offers zero or more products/services
CONSUMER	RESERVATION	1 : many	A consumer places zero or more reservations
PLATFORM_CHANNEL	RESERVATION	1 : many	A channel is used for zero or more reservations
RESERVATION	RESERVATION_ITEM	1 : many	A reservation contains one or more line items (mandatory)
PRODUCT_SERVICE	RESERVATION_ITEM	1 : many	A product appears in zero or more reservation items
RESERVATION	PAYMENT	1 : 1	A reservation is confirmed by exactly one payment
PAYMENT	TRANSACTION	1 : 1	A payment is recorded as exactly one transaction result


PROVIDER
int provider_id PK
string business_name
string contact_info
string address
string status
CONSUMER
int consumer_id PK
string full_name
string email
string phone
string address
PLATFORM_CHANNEL
int channel_id PK
string channel_name
string channel_type
string url
offers
places
made via
PRODUCT_SERVICE
int product_id PK
int provider_id FK
string name
string category
decimal price
string availability
string specifications
int stock_quantity
RESERVATION
int reservation_id PK
int consumer_id FK
int channel_id FK
date reservation_date
string status
timestamp created_at
contains
included in
confirmed by
RESERVATION_ITEM
int item_id PK
int reservation_id FK
int product_id FK
int quantity
decimal unit_price
PAYMENT
int payment_id PK
int reservation_id FK
string payment_method
decimal amount
string status
timestamp payment_timestamp
recorded as
TRANSACTION
int transaction_id PK
int payment_id FK
string transaction_ref
string result
timestamp processed_at

Task:2. Relational Model (3NF Normalized)

The relational model is derived from the conceptual model by converting entities into normalized tables. All tables satisfy First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF).

2.1 Normalization Analysis

First Normal Form (1NF)

- All attributes contain only atomic (indivisible) values.
- No repeating groups or multi-valued attributes exist.
- Each table has a clearly defined primary key.
- The specifications attribute in PRODUCT_SERVICE is stored as TEXT (a single atomic value per product).

Second Normal Form (2NF)

- All non-key attributes are fully functionally dependent on the entire primary key.
- RESERVATION_ITEM uses a surrogate PK (item_id), preventing partial dependencies on the composite (reservation_id, product_id).
- unit_price in RESERVATION_ITEM depends on both the reservation and the product at that moment — it is a derived snapshot, not a partial dependency.

Third Normal Form (3NF)

- No transitive dependencies exist between non-key attributes.
- Payment method details are stored in PAYMENT, not derived from or stored in RESERVATION.
- Transaction reference and gateway result are stored in TRANSACTION, not in PAYMENT.
- Provider business details do not appear in PRODUCT_SERVICE — they are accessed via FK join.

Key Design Decisions

- unit_price is stored in RESERVATION_ITEM as a historical snapshot. If the live price in PRODUCT_SERVICE changes, older reservations retain their original quoted price — preventing update anomalies.
- PAYMENT.reservation_id carries a UNIQUE constraint, enforcing the 1:1 relationship stated in the business rules.
- TRANSACTION is kept separate from PAYMENT to cleanly handle payment retries and gateway failures without overwriting the payment record.
- RESERVATION_ITEM acts as a mandatory bridge — a reservation with zero items violates the existence dependency rule (enforced at application level with a check before insert).

2.2 Table Schemas

Table 1: PROVIDER

Column	Type	Key	Description
provider_id	INT	PK	Auto-increment, unique identifier for each provider
business_name	VARCHAR(200)		Official registered name of the business, NOT NULL
email	VARCHAR(200)	UK	Unique login/contact email, NOT NULL
phone	VARCHAR(30)		Contact phone number
address	TEXT		Business address
status	ENUM		Account status: active | suspended | inactive
registered_at	TIMESTAMP		Account creation timestamp, DEFAULT NOW()

Table 2: PRODUCT_SERVICE

Column	Type	Key	Description
product_id	INT	PK	Auto-increment primary key
provider_id	INT	FK	References PROVIDER(provider_id), NOT NULL
name	VARCHAR(200)		Product or service name, NOT NULL
category	VARCHAR(100)		Classification category, NOT NULL
price	DECIMAL(12,2)		Current listed price, NOT NULL, CHECK >= 0
availability	ENUM		in_stock | out_of_stock | service_available
stock_quantity	INT		NULL for services; integer for physical goods
specifications	TEXT		Free-text description of product details
updated_at	TIMESTAMP		Last updated timestamp

Table 3: CONSUMER

Column	Type	Key	Description
consumer_id	INT	PK	Auto-increment primary key
full_name	VARCHAR(200)		Consumer's full name, NOT NULL
email	VARCHAR(200)	UK	Unique email address, NOT NULL
phone	VARCHAR(30)		Contact phone number
address	TEXT		Delivery/billing address
registered_at	TIMESTAMP		Registration timestamp

Table 4: PLATFORM_CHANNEL

Column	Type	Key	Description
channel_id	INT	PK	Auto-increment primary key
channel_name	VARCHAR(100)		Name of the channel (e.g., Main Website)
channel_type	ENUM		website | social_media | mobile_app
url	VARCHAR(300)		URL or handle of the channel

Table 5: RESERVATION

Column	Type	Key	Description
reservation_id	INT	PK	Auto-increment primary key
consumer_id	INT	FK	References CONSUMER(consumer_id), NOT NULL
channel_id	INT	FK	References PLATFORM_CHANNEL(channel_id), nullable
reservation_date	DATE		Date the reservation was made, NOT NULL
status	ENUM		pending | confirmed | cancelled | fulfilled
notes	TEXT		Optional consumer notes
created_at	TIMESTAMP		Record creation timestamp

Table 6: RESERVATION_ITEM

Column	Type	Key	Description
item_id	INT	PK	Auto-increment primary key
reservation_id	INT	FK	References RESERVATION(reservation_id), NOT NULL
product_id	INT	FK	References PRODUCT_SERVICE(product_id), NOT NULL
quantity	INT		Ordered quantity, NOT NULL, CHECK >= 1
unit_price	DECIMAL(12,2)		Price snapshot at reservation time, NOT NULL

Table 7: PAYMENT

Column	Type	Key	Description
payment_id	INT	PK	Auto-increment primary key
reservation_id	INT	FK+UK	References RESERVATION, UNIQUE (enforces 1:1), NOT NULL
payment_method	ENUM		cheque | electronic_card
amount	DECIMAL(12,2)		Total payment amount, NOT NULL, CHECK >= 0
status	ENUM		pending | completed | failed | refunded
payment_timestamp	TIMESTAMP		When payment was initiated

Table 8: TRANSACTION

Column	Type	Key	Description
transaction_id	INT	PK	Auto-increment primary key
payment_id	INT	FK+UK	References PAYMENT(payment_id), UNIQUE (1:1), NOT NULL
transaction_ref	VARCHAR(100)	UK	External payment gateway reference, UNIQUE
result	ENUM		success | failed | timeout
failure_reason	TEXT		Populated when result is failed or timeout
processed_at	TIMESTAMP		When gateway processed the transaction